DATA SCIENCE PORTFOLIO PROJECT

Child mortality and immunization trends in Bangladesh, 2000–2024

A reproducible statistical analysis with South Asian comparison

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October 2026

BACKGROUND

Bangladesh is a child survival success story. Is the progress continuing?

Q1 · Pace

How fast has child mortality fallen in Bangladesh, and has the decline slowed?

Q2 · SDG 3.2

Will Bangladesh reach under-5 ≤ 25 and neonatal ≤ 12 deaths per 1,000 by 2030?

Q3 · Determinants

How is mortality associated with immunization, income and water across South Asia?

Q4 · Prediction

Can a model predict child mortality in a country it has never seen?

DATA

Public UN and WHO data, compiled by the World Bank

Indicator	Unit	Original source
Under-5 mortality	per 1,000 live births	UN IGME
Infant mortality	per 1,000 live births	UN IGME
Neonatal mortality	per 1,000 live births	UN IGME
Measles immunization	% aged 12–23 months	WHO/UNICEF
DPT3 immunization	% aged 12–23 months	WHO/UNICEF
GDP per capita	constant 2015 US$	World Bank
Basic drinking water	% of population	WHO/UNICEF JMP

8 + 1

countries + South Asia region

25 years

2000–2024

200

country-years for modelling

DATA · RAW

The raw download: one value per row

data/raw/worldbank_indicators.csv

iso3	indicator	year	value
BGD	SH.DYN.MORT	2025	empty
BGD	SH.DYN.MORT	2024	30.5
BGD	SP.DYN.IMRT.IN	2024	24.4
BGD	SH.DYN.NMRT	2024	17.9
BGD	SH.IMM.MEAS	2024	96.0
BGD	SH.IMM.IDPT	2024	97.0
BGD	NY.GDP.PCAP.KD	2024	1,941.28
BGD	SH.H2O.BASW.ZS	2024	98.78

1,638 × 4

rows × columns

Long format: one Bangladesh-2024 record is split across 7 rows.

55 empty values, all for 2025, which is not yet published.

Indicators are coded, e.g. SH.DYN.MORT = under-5 mortality.

DATA · PROCESSED

After processing: one row per country-year

data/clean_panel.csv

Bangladesh rows shown · orange headings = columns created in processing

Year	U5MR	IMR	NMR	Measles	DPT3	GDP pc	Water	log U5MR	log GDP	Neonatal %
2000	85.5	59.8	43.4	74	82	620.56	94.56	4.45	6.43	50.76
2012	44.4	33.9	26.6	88	94	1,050.39	96.85	3.79	6.96	59.91
2024	30.5	24.4	17.9	96	97	1,941.28	98.78	3.42	7.57	58.69

Reshaped

1,638 rows became 225 rows × 17 columns

Trimmed

2025 dropped: no mortality data yet

3 new features

log U5MR, log GDP, neonatal share

4 quality flags

mark imputed values; all FALSE

METHODS

An end-to-end, reproducible pipeline

01
Acquisition
API download, retries, cached snapshot

02
Validation
Duplicates, ranges, consistency checks

03
Features
Log transforms, neonatal share, flags

04
Descriptive
Summary tables, ARR, figures

05
Statistics
Trend tests, SDG projection, panel models

06
Machine learning
Linear vs random forest, leave-one-out

07
Reporting
Summary, limitations, exported outputs

TOOLS

Python, pandas, statsmodels, scikit-learn, Kaggle

DATA QUALITY

Validated before any modelling

1,638

raw records

7 indicators × 9 areas × 26 years

0

duplicates

per country, indicator and year

0

failed checks

ranges, and neonatal ≤ infant ≤ under-5

0

values imputed

no gaps in 2000–2024

⚠ All 55 missing values were for 2025, which is not yet published, so the study window was trimmed to 2000–2024.

RESULTS · BANGLADESH

Under-5 mortality fell by almost two-thirds since 2000

85.5 → 30.5

under-5 deaths per 1,000 live births

−64%

reduction, 2000–2024

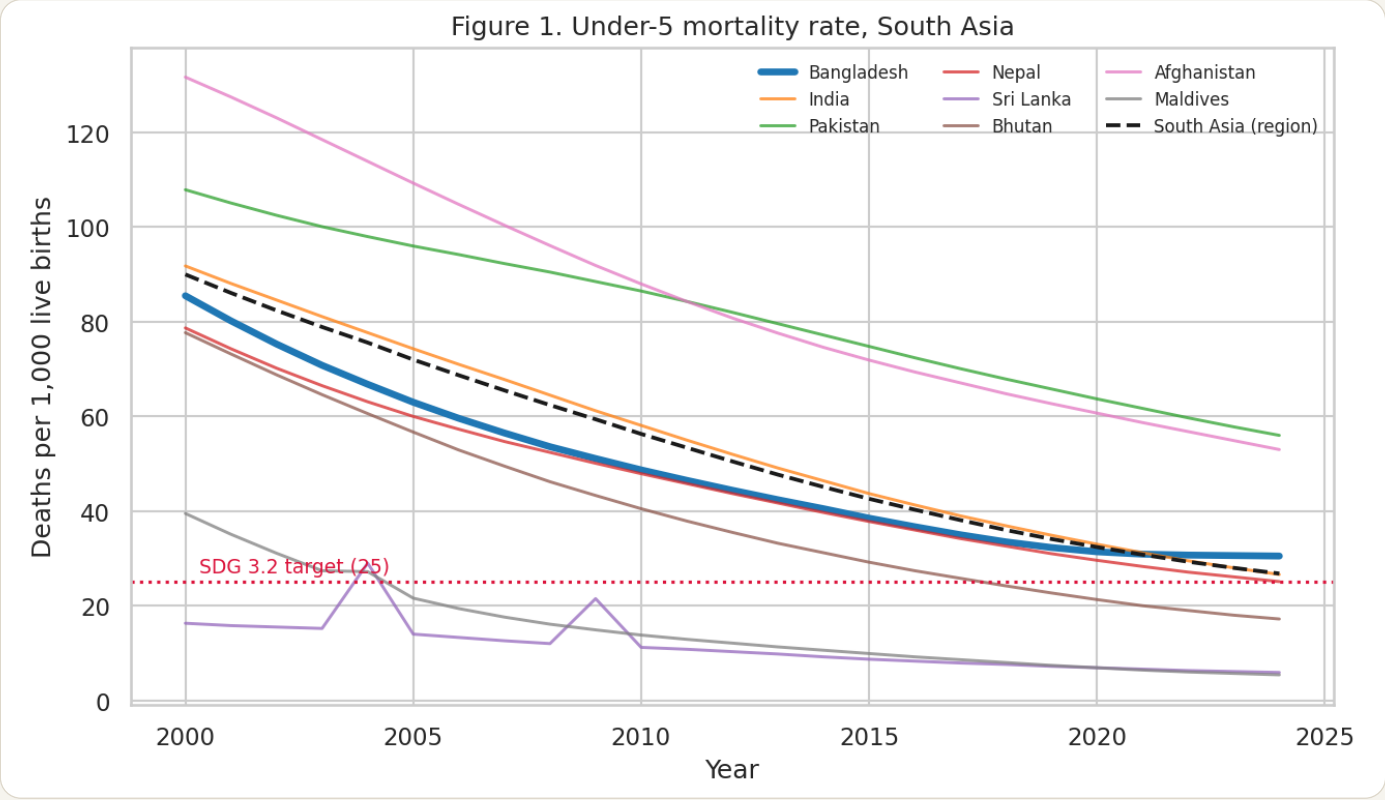
74% → 96%

measles immunization coverage

Year	Under-5	Infant	Neonatal	Measles %
2000	85.5	59.8	43.4	74
2010	48.7	36.8	28.8	88
2020	31.4	24.8	18.6	97
2024	30.5	24.4	17.9	96

RESULTS · SOUTH ASIA

Bangladesh has fallen behind India and Nepal



2000

Bangladesh (85.5) was below India and the regional average.

2024

Bangladesh (30.5) is above India (26.6), Nepal (25.1) and the region (26.8).

ONLY HIGHER

Pakistan (56.0) and Afghanistan (53.0).

RESULTS · PACE OF DECLINE

The decline slowed significantly after 2012

5.6% → **3.4%**

a year, 2000–2012

a year, 2012–2024

Slope change
p < 0.001

Annual rate of reduction	Before 2012	From 2012	Change in slope
Under-5 mortality	5.6%	3.4%	Significant, p < 0.001
Infant mortality	4.9%	3.0%	Significant, p < 0.001
Neonatal mortality	4.2%	3.7%	Not significant, p = 0.26

RESULTS · NEWBORNS

About 6 in 10 child deaths now happen in the first month of life

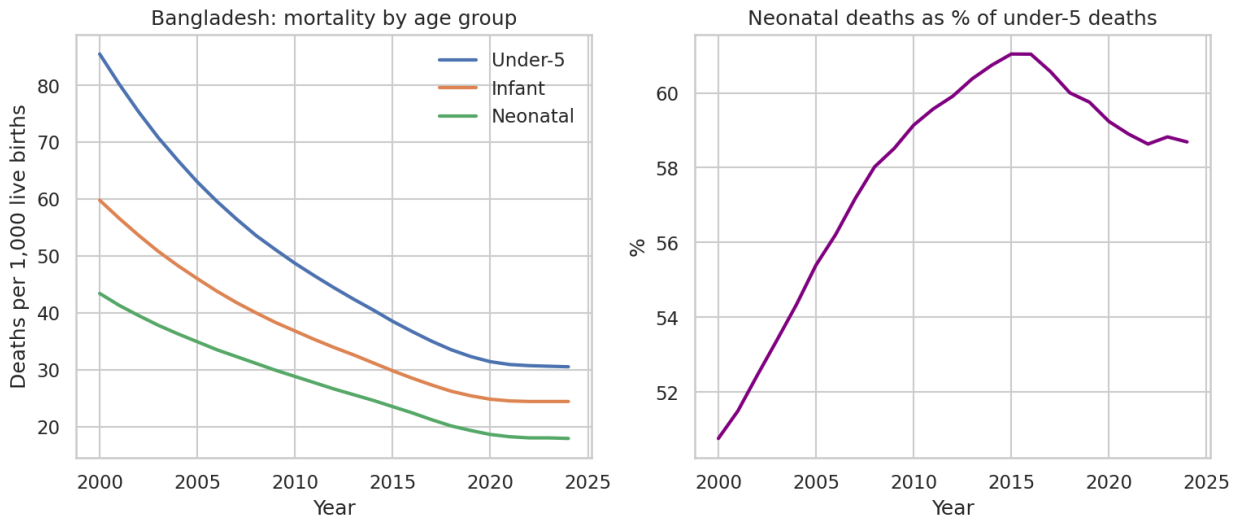
50.8% → 58.7%

share of under-5 deaths in the first 28 days, 2000 to 2024

3.9% vs 4.5%

yearly decline, neonatal vs under-5

Figure 2. Bangladesh: where child deaths are concentrated



RESULTS · SDG 3.2

Off track for newborns, borderline for under-5

Under-5 · target ≤ 25

Uncertain

Latest (2024)	**30.5**
Projected 2030 (95% CI)	**24.9 (24.0–25.8)**
Pace: current vs needed	**3.4% vs 3.3%**

Neonatal · target ≤ 12

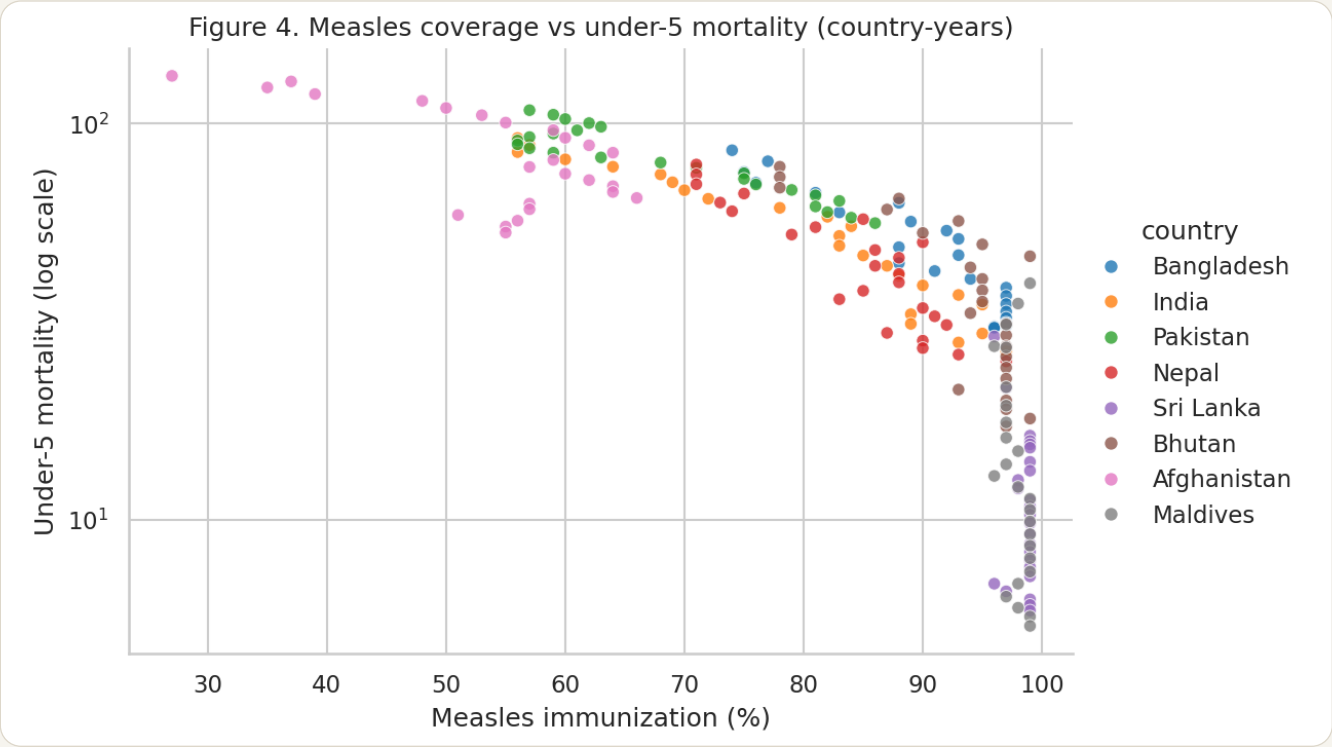
Off track

Latest (2024)	**17.9**
Projected 2030 (95% CI)	**14.3 (13.8–14.8)**
Pace: current vs needed	**3.7% vs 6.7%**

⚠ The projection assumes the 2012–2024 pace continues. At the 2020–2024 pace (about 0.7% a year), under-5 mortality would be about 29 in 2030, well above the target.

RESULTS · DETERMINANTS

Across countries, higher coverage goes with lower mortality



Each dot is one country-year: 8 countries, 2000–2024.

Low coverage (Afghanistan, Pakistan) comes with high mortality.

But at 96–99% coverage the dots form a **vertical stack**: mortality kept falling after coverage stopped rising.

RESULTS · DETERMINANTS

Within countries, the measles association disappears

Association with under-5 mortality	Pooled regression	Country fixed effects
Measles coverage, +10 points	−22.8% (p = 0.03)	+7.6% (p = 0.21)
log GDP per capita	−0.52 (p < 0.001)	−0.20 (p = 0.51)
Basic water access	+0.019 (p = 0.08)	+0.002 (p = 0.76)
Time trend, per year	−2.4% (p = 0.09)	−5.0% (p = 0.02)
R^2	0.82	0.98

The pooled result reflected differences **between** countries.

The decline continued from factors outside the model.

Child mortality in Bangladesh · 14 · Outcome: log under-5 mortality · 200 country-years · SEs clustered by country (8 clusters) · all VIFs below 4

MACHINE LEARNING

Testing prediction on countries the model has never seen

Round 1	AFG · test	BGD	BTN	IND	LKA	MDV	NPL	PAK
Round 2	AFG	BGD · test	BTN	IND	LKA	MDV	NPL	PAK
… Round 8	AFG	BGD	BTN	IND	LKA	MDV	NPL	PAK · test

Inputs → target

Measles, DPT3, log GDP, water, year → log under-5 mortality

Models

Linear regression (baseline) vs random forest (500 trees)

MACHINE LEARNING

Random forest wins overall, but both models fail out of range

LINEAR REGRESSION

0.63

log-RMSE · median error ~35%

RANDOM FOREST

0.55

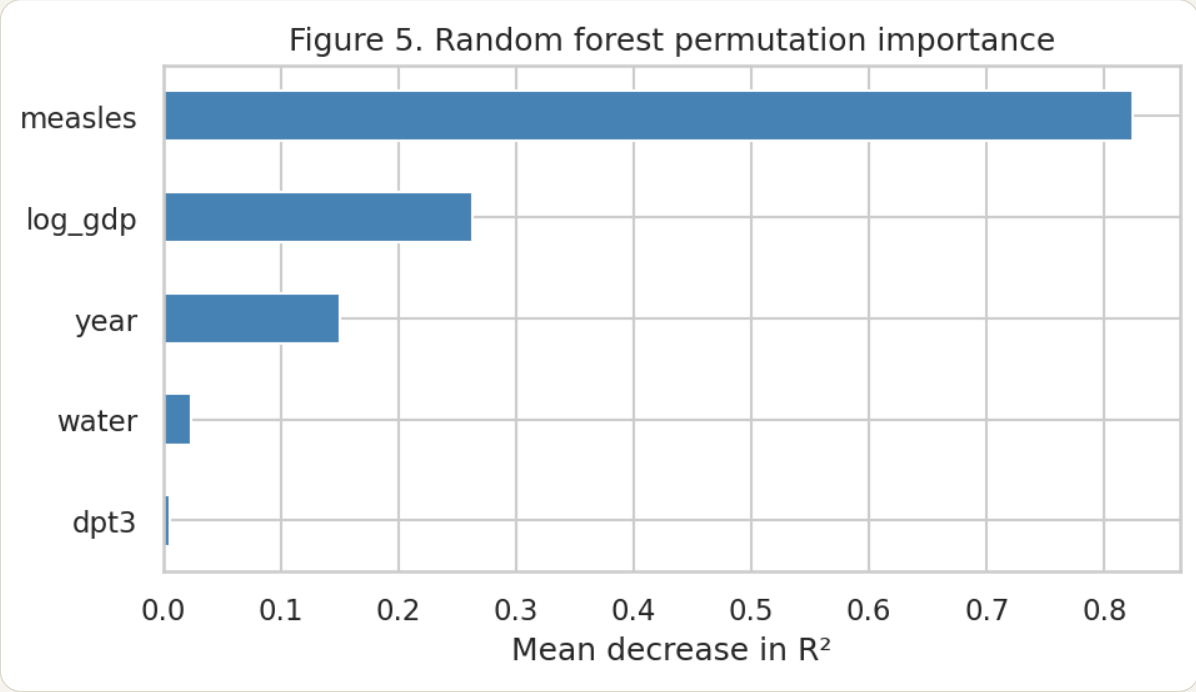
log-RMSE · median error ~24%

Held-out country	Linear	Forest	Why
Afghanistan	1.27	0.27	Coverage below the training range
Bangladesh	0.24	0.28	Mid-range: both work
Bhutan	0.42	0.64	
India	0.31	0.09	Forest near-exact
Maldives	0.44	0.50	
Nepal	0.29	0.29	
Pakistan	0.47	0.31	
Sri Lanka	0.85	1.19	Mortality below the training range

Child mortality in Bangladesh · 16 · Error in log under-5 mortality; lower is better; worst cases shaded

MACHINE LEARNING

Most predictive is not the same as most causal



Measles coverage is by far the strongest predictor.

Yet fixed effects found no within-country effect: it is a proxy for overall development.

DPT3 scores near zero only because it overlaps with measles.

MAIN MESSAGE

Bangladesh’s next gains in child survival must come from newborn care.

Progress slowed after 2012 and nearly stalled after 2020.

About 6 in 10 under-5 deaths happen in the first month.

Immunization has plateaued near 97% and cannot drive further gains alone.

LIMITATIONS

What this analysis cannot tell us

Modelled estimates

UN IGME figures are smoothed, so intervals understate uncertainty.

Country averages

Ecological data say nothing about individual children.

Association only

No result here is a causal effect.

Eight clusters

Cluster-robust p-values are approximate.

Fixed break year

2012 was chosen in advance; other years may differ.

Trend projection

Assumes the recent pace holds until 2030.

NEXT STEPS

From country trends to children at risk

1

Individual-level data

Use BDHS or MICS microdata to model child survival by maternal education, birth spacing and facility delivery.

2

Data-driven break points

Use joinpoint regression to estimate when the slowdown began, instead of fixing 2012.

3

Sub-national view

Compare divisions to find where progress stalled and where newborn care is weakest.

Thank you

Questions and feedback welcome

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Code and outputs: [https://www.kaggle.com/code/mdaddullah/child-mortality-bangladesh]